

EE 596H : Optimization Techniques

AY 2021-22 Winter Semester

Question 1

(5 marks)

Find points on the surface in $\mathbb{R}^3$ defined by $x_1^2 + 2x_2^2 + 3x_3^2 = 1$ where the tangent plane is parallel to the plane defined by $3x_1 - x_2 + 3x_3 = 1$.

Question 2

(7 marks)

Let $P \in \mathbb{R}^{n \times n}$ be a symmetric positive definite matrix. Show that

$$\lambda_{\min}(P) \|x\|^2 \leq x^T P x \leq \lambda_{\max}(P) \|x\|^2$$

Question 3

(7 marks)

In $\mathbb{R}^3$, consider the line which is formed by the intersection of the two planes defined by

$$\begin{aligned} 2x_1 + 2x_2 + 2x_3 &= 6 \\ 3x_1 + x_2 + 5x_3 &= 9 \end{aligned}$$

Find the point closest to the origin and is on the line by formulating an appropriate optimization problem.

Question 4

(7 marks)

Consider the constrained minimization problem

$$\begin{aligned} &\text{minimize} && f(x, y) = 2^{-(x+y)} \\ &\text{subject to} && 2^x + 2^y \leq 16 \\ &&& x \geq 0 \end{aligned}$$

Show that the problem above is a convex optimization problem. Write the KKT conditions for this problem and use them to determine the global minimum.

Question 5

(5 marks)

It is given that the function

$$f(x) = \frac{1}{x+2} - \frac{1}{x-5}$$

is unimodal in the interval $[0, 4]$. Consider three iterations of the Interval Halving method.

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- (a) List the sequence of points $x_1, x_2, \dots$ used.
 - (b) List the function values $f(x_1), f(x_2), \dots$ used.
 - (c) List the intervals obtained after each iteration.
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Question 6

(5 marks)

Consider the function $f(x) = x^3 - 6x$ where $x \in \mathbb{R}$.

- (a) Starting with $x^{(0)} = 1$, compute 2 steps of secant method.
 - (b) Starting with $x^{(0)} = -1$, compute 2 steps of secant method.
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Question 7

(7 marks)

Consider the problem of finding the minimizer of the function

$$f(x) = x_1^2 - 2x_1x_2 + 3x_2^2$$

in $\mathbb{R}^2$.

- (a) Compute the gradient and Hessian of f .
 - (b) Apply steepest descent method starting with $x^{(0)} = [1, 1]^T$. Compute $x^{(1)}$ and $x^{(2)}$.
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Question 8

(7 marks)

Consider the problem of finding the minimizer of the function

$$f(\mathbf{x}) = x_1^4 + 2x_1^2x_2^2 + x_2^4$$

in $\mathbb{R}^2$.

- (a) Compute the gradient and Hessian of f .
 - (b) Apply Newton's method starting with $x^{(0)} = [1, 1]^T$. Compute $x^{(1)}$ and $x^{(2)}$.
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